

# Joel Mugenyi

New York, NY, USA



+1 (917) 392-7312



mugenyij@gmail.com



joelmugenyi



<https://joelmugenyi.com/>

Energy systems researcher and PhD candidate specializing in electricity demand, energy affordability, and household consumption modeling. Experienced in econometrics, data science, and policy-relevant analysis using large data.

## Experience

- 2022 – present  **Research Assistant:** [Quadracci Sust. Energy Lab](#), Columbia University (NYC, USA).
- 2024  **Energy Modelling Consultant:** [Equatorial Power](#), Kampala, Uganda.
-  **Energy Fellow:** [Center on Global Energy Policy](#), Columbia University (NYC, USA).
- 2023  **Grid Reliability Consultant:** [Electricity Regulatory Authority](#), Kampala, Uganda.
- 2019 – 2022  **Research Engineer:** [eGUIDE Initiative](#), UMass-Amherst (Amherst, USA).
- 2018 – 2019  **R&D Associate:** [PowerGen Renewable Energy](#), Nairobi, Kenya.
- 2017  **Maintenance Engineer:** [Eskom Generation Limited](#), Jinja, Uganda.

## Education

- 2022 – 2026\*  **Ph.D., Earth & Environmental Engineering** Columbia University  
*Thesis: Building Affordable Energy Systems in Low-Income Communities*
- 2017 – 2019  **M.Sc., Electrical & Computer Engineering** Carnegie Mellon University  
Concentrations: *Data Science; Decentralized Energy Systems*
- 2012 – 2016  **B.Sc., Electrical Engineering** Makerere University  
Specializations: *Power Systems; Machines; Protection*

## Teaching

- Fall 2024  **Teaching Assistant:** Solid & Hazardous Waste Management, Columbia University.
- Fall 2018  **Teaching Assistant:** Photovoltaic Systems Engineering, Carnegie Mellon University.
- Spring 2018  **Teaching Assistant:** Software Defined Radios, Carnegie Mellon University.

## Research Publications

### Journal Articles

- 1 **J. Mugenyi** and V. Modi, “Survey bias and the poor: How survey responses overstate electricity spending,” *Energy for Sustainable Development*, vol. 90, p. 101 876, ISSN: 0973-0826.  DOI: [10.1016/j.esd.2025.101876](https://doi.org/10.1016/j.esd.2025.101876).
- 2 L. Masselus, J. Ankel-Peters, G. Gonzalez Sutil, V. Modi, **J. Mugenyi**, A. Munyehirwe, N. Williams, and M. Sievert, “Adoption of electricity in rural rwanda 10 years after connection,” *Nature Communications*, vol. 16, no. 1, p. 10 942, Dec. 7, 2025, Publisher: Nature Publishing Group, ISSN: 2041-1723.  DOI: [10.1038/s41467-025-66986-0](https://doi.org/10.1038/s41467-025-66986-0).

\*Expected graduation.

- 3 **J. Mugenyi**, G. G. Sutil, and V. Modi, “Electricity consumption: The role of grid reliability in appliance ownership and usage in rwanda,” *Energy Economics*, vol. 151, p. 108 907, 2025, ISSN: 0140-9883.  DOI: <https://doi.org/10.1016/j.eneco.2025.108907>.
- 4 **J. Mugenyi**, B. Muhwezi, S. Fobi, C. Massa, J. Taneja, N. J. Williams, and V. Modi, “Post-connection electricity demand and pricing in newly electrified households: Insights from a large-scale dataset in rwanda,” *Energy Policy*, vol. 198, p. 114 449, Mar. 1, 2025, ISSN: 0301-4215.  DOI: [10.1016/j.enpol.2024.114449](https://doi.org/10.1016/j.enpol.2024.114449).

## Conference Proceedings

- 1 **J. Mugenyi**, H. Natuhweera, and V. Modi, “Patterns, purchases, and policy: Understanding domestic electricity use in rural and peri-urban uganda,” presented at the 2025 Africa Electricity Symposium.
- 2 **J. Mugenyi**, I. V. Kinhonhi, and V. Modi, “Electricity outages in uganda: Causes, trends and regional disparities,” in *2024 IEEE PES/IAS PowerAfrica*, Oct. 2024, pp. 1–5.  DOI: [10.1109/PowerAfrica61624.2024.10759464](https://doi.org/10.1109/PowerAfrica61624.2024.10759464).
- 3 **J. Mugenyi**, I. V. Kinhonhi, and V. Modi, “Natural language processing reveals core issues in uganda’s power grid: A study of outages from 2015-2022,” in *2024 IEEE Global Humanitarian Technology Conference (GHTC)*, ISSN: 2473-5728, Oct. 2024, pp. 137–142.  DOI: [10.1109/GHTC62424.2024.10771564](https://doi.org/10.1109/GHTC62424.2024.10771564).
- 4 S. Fobi, **J. Mugenyi**, N. J. Williams, V. Modi, and J. Taneja, “Predicting levels of household electricity consumption in low-access settings,” in *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Waikoloa, HI, USA: IEEE, Jan. 2022, pp. 2213–2222, ISBN: 978-1-6654-0915-5.  DOI: [10.1109/WACV51458.2022.00227](https://doi.org/10.1109/WACV51458.2022.00227).
- 5 **J. Mugenyi**, E. Mugume, N. J. Williams, J. Kimani, K. Campbell, E. Uche, and J. Dougherty, “Smart metering technologies for mini grids in africa: An overview,” in *2021 IEEE PES/IAS PowerAfrica*, Aug. 2021, pp. 1–5.  DOI: [10.1109/PowerAfrica52236.2021.9543294](https://doi.org/10.1109/PowerAfrica52236.2021.9543294).

## Skills

|                 |                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming     |  Python, R, Java, SQL, L <sup>A</sup> T <sub>E</sub> X.                           |
| Databases       |  MySQL, PostgreSQL, SQLite.                                                       |
| Remote Sensing  |  QGIS, ArcGIS, ENVI.                                                              |
| Cloud Computing |  AWS (Lambda, EC2, S3, API Gateway)                                               |
| Power Systems   |  DIGSILENT PowerFactory, HOMER                                                    |
| Methods         |  Causal inference, regression analysis, panel data methods, machine learning      |
| Professional    |  Technical writing, teaching, stakeholder engagement, and research communication. |